

WILLIS ISLAND, Australia

WMO: 942990

Lat: 16.288S

Lon: 149.965E

Elev: 32

StdP: 14.68

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	70.6	71.5	53.4	60.8	72.7	55.6	65.9	73.6	32.2	73.4	30.9	74.1	18.7	160	0.755

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.2	88.7	79.5	87.8	79.1	87.0	78.7	81.5	86.0	80.8	85.5	80.2	84.8	11.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.1	156.5	84.4	79.3	152.4	84.1	78.8	149.8	83.9	45.3	86.5	44.5	85.8	43.9	85.2	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
31.3	29.3	27.6	DB	68.5	90.2	1.2	1.2	67.7	91.1	67.0	91.8	66.3	92.4	65.4	93.3
			WB	58.6	82.8	1.4	0.7	57.6	83.3	56.8	83.7	56.0	84.1	55.0	84.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	79.7	83.2	83.3	82.2	80.8	78.8	76.8	75.7	75.6	77.1	79.2	81.1	82.9	(6)
	DBStd	3.25	1.63	1.70	1.80	1.48	1.55	1.82	1.77	1.63	1.56	1.54	1.43	1.69	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD50	10842	1030	931	998	924	894	804	796	792	813	905	934	1021	(10)
	CDD65	5367	565	511	533	474	429	354	331	327	363	440	484	556	(11)
	CDH74	46927	6607	6013	5905	4669	3348	1901	1226	1098	1863	3373	4672	6252	(12)
	CDH80	10345	2219	2054	1629	764	217	41	11	13	83	415	950	1949	(13)
(14) Wind	WSAvg	17.9	14.9	14.2	17.2	20.0	20.4	20.7	20.6	19.7	18.1	17.7	16.7	15.0	(14)
(15) Precipitation	PrecAvg	40.1	7.4	7.9	7.4	4.4	2.9	1.9	1.3	0.6	0.4	0.6	1.5	4.0	(15)
	PrecMax	65.7	25.1	21.6	22.9	15.4	14.1	7.8	8.7	3.0	2.6	3.4	14.0	14.2	(16)
	PrecMin	9.5	0.5	0.4	0.3	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	12.9	6.1	5.8	5.6	3.3	3.1	1.9	1.4	0.6	0.6	0.8	2.3	3.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.8	89.7	88.5	85.9	84.1	82.5	81.2	81.5	83.1	85.5	87.6	89.6	(19)
		MCWB	80.1	80.0	79.9	78.8	76.7	75.3	73.5	73.3	75.0	76.7	78.3	79.7	(20)
	2%	DB	88.5	88.2	87.2	84.9	82.8	81.2	80.1	80.1	81.9	84.4	86.1	88.2	(21)
		MCWB	79.5	79.6	79.5	77.9	75.4	73.8	72.4	72.0	73.5	75.9	77.1	78.8	(22)
	5%	DB	87.5	87.3	86.2	84.2	82.0	80.2	79.1	79.2	81.0	83.4	85.2	87.3	(23)
		MCWB	78.9	79.3	78.9	77.1	74.7	72.9	71.6	70.8	72.3	75.0	76.3	78.2	(24)
	10%	DB	86.4	86.5	85.2	83.4	81.3	79.3	78.3	78.2	80.1	82.4	84.3	86.4	(25)
		MCWB	78.3	79.1	78.4	76.6	74.1	72.4	70.9	70.1	71.6	73.9	75.6	77.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.9	82.3	81.8	82.8	78.9	77.5	76.2	77.6	77.3	78.7	80.9	81.5	(27)
		MCDB	87.1	87.3	86.3	83.4	81.9	80.5	79.2	78.4	80.8	83.5	84.3	87.0	(28)
	2%	WB	81.0	81.4	80.8	80.1	77.4	76.0	74.7	75.2	75.6	77.5	79.2	80.5	(29)
		MCDB	86.1	86.3	85.4	82.7	81.2	79.5	78.5	78.2	79.8	82.1	83.7	86.0	(30)
	5%	WB	80.4	80.7	80.0	79.0	76.5	74.9	73.6	73.5	74.5	76.6	78.3	79.8	(31)
		MCDB	85.4	85.6	84.5	82.5	80.6	78.7	77.9	77.5	79.3	81.3	83.0	85.2	(32)
	10%	WB	79.9	80.0	79.4	78.2	75.6	74.1	72.7	72.2	73.4	75.7	77.3	79.1	(33)
		MCDB	84.7	84.8	83.7	82.0	80.0	78.3	77.3	76.8	78.6	80.7	82.3	84.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	6.2	5.6	5.1	4.7	4.7	4.9	5.4	6.1	6.5	6.7	6.8	(35)
		MCDBR	7.3	6.9	6.3	5.5	5.1	5.1	5.3	6.0	6.6	7.1	7.3	7.4	(36)
	5% WB	MCWBR	3.5	3.7	3.5	3.6	3.4	3.4	3.5	3.8	3.7	3.4	3.4	3.4	(37)
		MCWBR	6.6	6.5	6.0	5.4	4.7	4.9	4.9	5.4	6.2	6.7	7.0	6.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.387	0.389	0.395	0.390	0.377	0.378	0.360	0.364	0.373	0.388	0.394	0.386	(40)
		taud	2.614	2.623	2.590	2.583	2.583	2.574	2.597	2.578	2.533	2.510	2.524	2.587	(41)
	Ebn at Noon		304	300	292	282	275	269	278	286	293	297	299	304	(42)
		Edh at Noon	33	32	32	30	29	28	28	30	34	36	36	34	(43)
(44) All-Sky Solar Radiation	RadAvg		2078	2019	1801	1674	1469	1316	1423	1696	2023	2220	2284	2265	(44)
	RadStd		193	186	144	122	108	99	80	108	68	123	171	195	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page